ECN711
Galina Vereshchagina

Homework 3

1. Consider the *cake-eating problem*: Suppose that an infinitely-lived consumer has a cake of size w_0 in period 0 and needs to decide how much of it, c_t to eat in every period. The cake does not depreciate or grow, implying that the consumer is constrained by

$$c_t + w_{t+1} \leq w_t, \quad w_{t+1} \geq 0, \quad t \geq 0.$$

The consumer's per-period utility is given by the CRRA function, $u(c) = \frac{c^{1-\sigma}-1}{1-\sigma}$, and the consumer discounts future with the discount factor $\beta \in (0, 1)$.

   (a) (5 pts) Set up the consumer's decision problem.

   (b) (5 pts) Does this decision problem have a steady state? Provide a formal argument and explain intuitively.

   (c) (10 pts) Using the 'guess and verify' approach, solve for the agent's optimal choice of cake size and consumption. Is your answer consistent with the solution of the optimal growth model for $u(c) = log(c)$ and $f(k) = Rk$ found in class?

2. Consider the infinite horizon optimal growth model with $u(c) = log(c)$ and $f(k) = Ak^\alpha + (1-\delta)k$, where $A > 0$, $\alpha, \delta \in (0, 1)$ and k_0 is given.

   (a) (10 pts) What approach would you use to solve for the optimal path of capital? Either obtain the closed form solution, or explain why you are not able to do it.

   (b) (10 pts) Derive the condition that determines the steady state level of capital k^*.

   (c) (10 pts) How is the steady state capital level affected if A increases? depreciation δ rises? Provide formal arguments and intuitive explanations.

   (d) (20 pts) Modify your shooting algorithm to compute the optimal capital profile that starts from a given k_0 and converges to the steady state level k^*.
   Suppose that $A = 1$, $\alpha = 0.4$, $\delta = 0.1$, $\beta = 0.96$ and the economy is in the steady state (i.e. $k_0 = k^*$). Now suppose that A increases to 1.1. Compute and plot how this change in production technology affects the optimal path of capital. Print out your plots, attach them to the hard copy of the HW, and send the codes to the teaching assistant.

3. Consider an infinite horizon optimal growth model with CRRA per period utility function $u(c) = \frac{c^{1-\sigma}-1}{1-\sigma}$ and $f(k) = Ak^\alpha$.

   (a) (5 pts) Set up the sequential problem.

   (b) (10 pts) Write down the recursive formulation of this problem, clearly define state and control variables and explain what are the component parts of the solution to this problem.

   (c) Assume that $\sigma = 1$. Show that the solution to the sequential problem also solves the dynamic programming problem. To do this, you need to:

- (10 pts) find the solution of the sequential problem (you can refer to any of the findings from our classwork or your previous homeworks), and find an expression for the indirect life-time utility function $V(k_0)$ obtained after we solve the sequential problem;
- (5 pts) find $g(k)$ which maximizes the expression under $\max$ in the Bellman equation, in which $V(\cdot)$ is the indirect utility function obtained from the sequential problem. How does $g(k)$ relate to the capital path $\{k^*_{t+1}\}_{t=0}^{\infty}$ that solves the sequential problem?
- (5 pts) show that the indirect life-time utility function $V(\cdot)$ indeed solves the Bellman equation.

4. **Optimal growth model with fixed income**
Consider an infinite horizon optimal growth model, in which the agent receives an exogenous flow of income $\{y_t\}_{t=0}^{+\infty}$ in addition to the (endogenous) capital income $f(k)$. Suppose that the income in the period $t+1$ can be expressed as function of the income in period t, $y_{t+1} = h(y_t)$. For example, there might be some seasonal fluctuations in income: $h(y_t) = 1$ if $y_t = 0$ (winter in period t) and $h(y_t) = 0$ if $y_t = 1$ (summer in period t).

   (a) (10 pts) Write down the recursive formulation of this problem. Carefully define all state and control variables, list all the component parts of the solution.

   (b) (10 pts) Now assume that the sequence of fixed income is described by the following law: $y_{t+1} = h(y_t, y_{t-1})$. Reformulate the recursive problem.